ACROPOLIS INSTITUTE OF TECHNOLOGY AND RESEARCH

Department of Information Technology

Synopsis

on

PLANT DISEASE PREDICTION USING SYMPTOMS ANALYSIS

1. INTRODUCTION

1.1 Problem Context

Agriculture plays a major role in the economy, but plant diseases continue to cause major crop losses every year. Farmers often fail to identify diseases in the early stage due to lack of technical knowledge, delayed expert consultation, and limited access to agricultural support in rural areas.

- Requires trained specialists
- Consumes more time
- Is difficult to access in remote regions
- Does not provide quick preliminary assistance
- Difficult to monitor large agricultural fields manually
- Manual diagnosis may lead to human error

Plant diseases spread rapidly and reduce both crop quality and productivity. Traditional disease detection methods mainly depend on manual inspection by agricultural experts, which:

In many rural farming areas, farmers rely mainly on traditional methods and personal experience to identify plant diseases. Due to the absence of immediate expert guidance, diseases are often recognized only after severe damage has already affected the crops. Environmental factors such as changing climate conditions, soil infections, and pest attacks further increase the risk of plant diseases and make manual identification more difficult. As a result, farmers may use incorrect pesticides or treatments, leading to additional financial losses and reduced agricultural productivity.

Therefore, there is a need for an intelligent, low-cost, and accessible plant disease prediction system that can assist farmers using automated symptom analysis.

1.2 Motivation & Need

Most existing plant disease detection systems focus mainly on image classification and often ignore symptom descriptions provided by farmers. Many solutions also require high-quality images and internet connectivity, which may not always be available in rural areas.

The major limitations of current systems are:

- High dependency on agricultural experts
- Delayed disease identification
- Reduced accuracy with poor image quality
- Limited accessibility in rural regions
- Lack of symptom-based prediction systems
- Lack of user-friendly interfaces for non-technical farmers
- Time-consuming manual monitoring of large agricultural fields
- Low awareness and adoption of AI-based agricultural technologies in rural areas
- Limited support for multiple crop diseases
- Lack of real-time disease prediction support

he motivation behind this project is to develop a smart and user-friendly system capable of:

- Detecting plant diseases at an early stage
- Using symptom descriptions for prediction
- Providing faster preliminary analysis
- Supporting farmers with accessible AI-based assistance
- Reducing crop losses through timely disease identification

- Increasing plant diseases are affecting crop productivity and food supply worldwide.
- Small-scale farmers often cannot afford expert agricultural consultation.
- Early disease prediction can help reduce unnecessary pesticide usage.
- Manual plant disease monitoring is time-consuming for large farms.
- Automated disease detection reduces dependency on manual monitoring.

2. LITERATURE SURVEY

TABLE I

PREVIOUS YEAR RESEARCH PAPER

Year	Authors	Title	Key Findings
2020	Liakos, K.G. et al	Deep Learning Models for Plant Disease Detection and Diagnosis	Deep learning and CNN models achieved very high accuracy in automatic plant disease detection.
2021	Sharma, R. et al.	Symptom-Based Plant Disease Prediction Using Naive Bayes Algorithm	Symptom-based prediction provided faster preliminary disease analysis without image dependency.
2022	Chen, L. et al.	Machine Learning in Agriculture: A Review	Reviewed the importance of AI, ML, and automation in improving smart agriculture systems.
2023	Gupta, R. & Jain, M.	Hybrid CNN Model for Plant Disease Classification	Hybrid CNN models improved classification accuracy and minimized false disease detection.

Existing Applications

TABLE II
EXISTING APPLICATIONS/PRODUCTS

Application/Product	Purpose	Availability
Plantix	Plant disease detection using crop symptoms	Mobile App
Leaf Doctor	Plant health analysis	Mobile App
AgroAI	Smart farming assistance	Web/Mobile

Technologies Currently Used

- Artificial Intelligence (AI)
- Deep Learning
- Convolutional Neural Networks (CNN)
- TensorFlow/Keras
- OpenCV
- Scikit-learn

2.1 Problem Statement

To develop an AI-based plant disease prediction system capable of identifying

selected plant diseases using symptom analysis techniques with improved accessibility, faster response time, and better preliminary prediction accuracy.

3. PROPOSED SOLUTION OVERVIEW

The proposed system is an AI-powered web application that accepts symptom descriptions related to plant conditions such as leaf colour changes, spots, wilting, dryness, or fungal growth.

The proposed system is designed as an AI-based web application that helps farmers predict plant diseases using symptom analysis techniques. Users can enter symptoms related to plant conditions such as leaf colour changes, yellowing, spots, wilting, dryness, curling, or fungal infections through a simple and user-friendly interface. The system analyzes these symptoms using machine learning algorithms trained on plant disease datasets to identify patterns associated with different plant diseases. The final result displays:

- Predicted plant disease
- Confidence score
- Basic preventive measures and treatment suggestions

4. OBJECTIVES

1. To develop a web-based plant disease prediction system using AI techniques.
2. To predict selected plant diseases using symptom analysis.
3. To improve prediction accuracy using machine learning algorithms.

5. CONTRIBUTIONS

The proposed project introduces several improvements over traditional systems:

- Symptom-based plant disease prediction
- Lightweight and user-friendly web interface
- Faster preliminary predictions using machine learning
- Reduced dependency on agricultural experts for initial diagnosis`

6. THEORETICAL ANALYSIS

6.1 Block Diagram

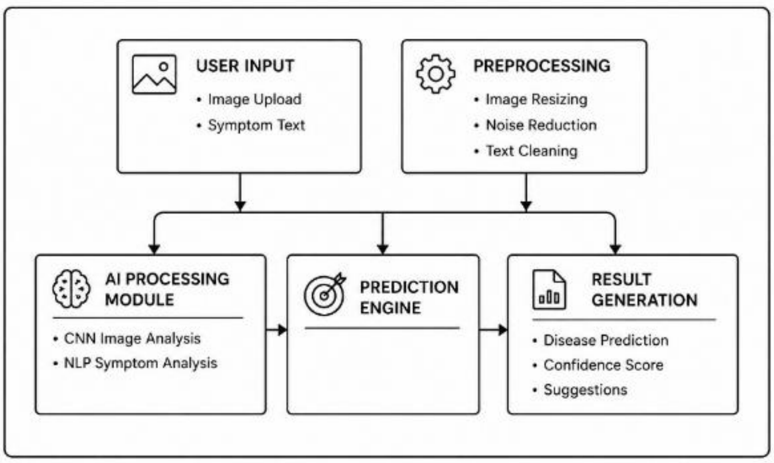

Fig. 1: Plant and Disease Prediction Diagram

6.2 Hardware / Software Requirements

Software Requirements

TABLE III
SOFTWARE REQUIREMENTS

Software	Purpose
Python	Backend programming
Scikit-learn	Machine learning
Flask	Web framework
HTML/CSS/JavaScript	Frontend
VS Code	Development environment
MySQL/Firebase	Database

TABLE IV
Hardware Requirements

Component	Specification
Processor	Intel i5 or higher
RAM	Minimum 8 GB
Storage	256 GB SSD
Camera	Optional for live capture
GPU	Optional for faster training

7.APPLICATIONS

- Smart Agriculture
- Crop Health Monitoring
- Rural Farming Assistance
- AI-based Farming Support Systems
- Agricultural Research Assistance
- Sustainable Farming Practices
- Agricultural Advisory Platforms
- Crop Yield Improvement Programs

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Link of GitHub Repository:

https://github.com/sudarshanr04/EARLY-DISEASE-DETECTION.git